◆ CZTVN · DATA INFRASTRUCTURE · FIELD MANUAL

The Data Ladder

A field manual for the CZTVN traffic-data stack — from free-tier mastery, to the dashboard that proves it, to the first transaction.

Every rung earns the next. This one documents all three, assumes a beginner at the keyboard, and marks the win at each step.

CZTVN-DATA-2026-0711

Print Edition · July 11, 2026 · Shannon, Georgia

DOCUMENT CONTROL

A Grade A1 record of the CZTVN data stack

DOCUMENT ID	CZTVN-DATA-2026-0711
SERIES	The CZTVN Newsdesk — Data Infrastructure
COMPANION TO	cztvn-source-hub.html (newsdesk pull) & cztvn-data-infrastructure-hub.html (the ladder)
SUBJECT	The traffic-data resource stack, its access tiers, and the CZTVN dashboard build
ENTITY OF RECORD	TM CZAR Television Network, Inc. (CZTVN) · cross-references DJWZ Coding / 859SSCS1
DATE	July 11, 2026
LENGTH REGISTER	Exhaustive
REGISTER	Field manual + implementation guide + build-spec
READER STATUS	Beginner assumed — every technical step is walked, every win is marked
CLASSIFICATION	Internal record · Grade A1
AUTHOR	Josh Wayne — Founder / Architect

HOW TO READ THIS MANUAL

Section I frames the ladder and the ownership thesis behind it. Sections II–IV walk the three rungs — free/dev-tier, agency, and the paid milestone — exhaustively, each resource with beginner build-steps. Section V is the dashboard build-spec: the architecture you commission, in plain terms, in the order to build it. Section VI is the 859SSCS1 contribution loop that turns the build into public standing. Section VII is the Build Register and the operator's own words. Throughout, a **Eureka** mark celebrates each milestone — the first key, the first call, the first city read, the first transaction.

THE INSTRUMENTS

The recurring devices in this manual

Five marks repeat through the sections. Learning them once makes the rest scan fast.

FREE DEV-KEY AGENCY PAID The **access tier** — open, free-with-a-key, partnership-gated, or costs money.

OWNABLE RENTED GATED The **ownable flag** — whether the terms let you keep and reprocess the data, or only access it. This is the flag that decides what can legally feed the dashboard.

1 A **build-step** — one concrete, beginner-level action. The agent can do the typing; the step is what you are directing it to do.

◆ A **Eureka** — a milestone worth feeling. Each one is a small proof that the stack is real and yours.

> *Master the free resources — and demonstrate that knowledge through the dashboard.*
>
> THE OPERATOR, ON THE INTENT OF THIS MANUAL

CONTENTS

Seven sections, with full-page dividers

THE GITHUB LAB · FIELD MANUAL · SECTION

I

The Ladder

The three rungs, and the ownership thesis behind them

3	1	OWN→
RUNGS	THESIS	NOT RENT

I.1 Two Hubs, One Stack

The newsdesk hub pulls the news; it clears before a show. This manual documents the layer beneath it — the data infrastructure — and it is shaped as a ladder rather than a flat list, because the stack has a natural order of arrival. You reach the rungs in sequence, and each one earns the next.

Rung One is free and dev-tier: everything you can touch today, at no cost, and the surface the dashboard is built on. Rung Two is agency access — real feeds you cannot open solo yet, worth knowing now so you can partner into them later. Rung Three is the paid rung: where the entity opens its account and spends for the first time.

I.2 The Ownership Thesis

The single idea that organizes the whole ladder: **access is not ownership.** Renting a data feed gives you more calls; it never gives you an asset. The commercial APIs will happily sell more access, but their terms forbid storing or redistributing what comes back. The resources that move CZTVN toward a proprietary asset are the ones you can keep — a licensed dataset, a free government feed, and above all your own capture. That distinction is printed on every card in this manual as the *ownable* flag, and it is what decides which sources may legally feed the dashboard.

> *Access is the fast layer you rent. Ownership is the durable layer you build. The ladder points at the second one.*

I.3 You Are Here

This manual assumes no prior work with APIs or data infrastructure. That is deliberate and it is not a limitation. The model the whole practice runs on is the operator who directs the work rather than types every line — you name the goal and the standard, an agent does the building, and you own the record. So each build-step below is written as an instruction you can hand off, and each is small enough to finish in one sitting.

The reason for the Eureka marks is the same. A beginner building real infrastructure deserves to feel the moments that matter: the first time a road segment reports its own speed back to you, the first time two cities line up on the same yardstick, the first time money moves for something you will keep. Those are not decoration. They are proof, dated and felt, that the stack is becoming real — and they are the same celebrate-by-default instinct the network runs on.

RUNG	WHAT IT IS	WHAT YOU DO HERE
One · Free & Dev-tier	Open + key-gated sources, no spend	Master them; build the dashboard on them
Two · Agency	Partnership-gated public feeds	Learn them; qualify and partner later
Three · Paid	Licensed data & premium access	Open the account; buy what you can own

◆ EUREKA

Reaching the end of this page is the first one. You now hold the whole map of the stack in your head — the three rungs, and the one thesis that ranks everything on them.

THE GITHUB LAB · FIELD MANUAL · SECTION

II

Rung One

Free & Dev-Tier — the training ground

12
SOURCES

$0
SPEND

BUILD
SURFACE

II.0 The Training Ground Is Also the Build Surface

Everything on this rung is reachable today at no cost. Some sources are wholly open; others are free developer tiers that need an API key — still no spend, just a sign-up. This is where you master the stack, and it is also where the dashboard gets built, because the dashboard runs on exactly these dev-tier feeds.

◆ 859SSCS1 · THE OPEN-SOURCE LEDGER

The dashboard you assemble on this rung is itself an artifact under the **github.com/859SSCS1** vault — a public, dated, build-in-public demonstration of API competence. It runs on the ledger's own loop:

`STUDY the gap → IMPLEMENT via Claude Code → CERTIFY the published build → log to the Build Register.`

The agent writes the code; you direct it and own the record. So ‘master the free resources’ and ‘demonstrate that mastery in the dashboard’ are not two goals — they are one line in the ledger.

RUNG ONE HOLDS FOUR REAL-TIME APIS, TWO CITY-CYCLE BENCHMARKS, THREE FREE GOVERNMENT FEEDS, AND THREE STRUCTURAL DATASETS.

II.1 Real-Time Flow & Incidents

Four commercial APIs report what traffic is doing right now — observed speeds, travel times, and incidents. All four offer a free developer tier. All four rent access rather than sell ownership, so they are the dashboard's live layer, not its archive. Start with one; the others follow the same shape.

TomTom Traffic API

DEV-KEY

RENTED

Live flow — observed speed versus the road's free-flow speed — plus incidents like closures, construction, and crashes, refreshed roughly every 30 to 60 seconds. The friendliest first key in the set.

`→ dashboard: the live congestion + incident layer on the map`

FIRST CONTACT — TOMTOM, STEP BY STEP

1. Create a free account at `developer.tomtom.com` and confirm your email.
2. In the dashboard, create an application — this generates your **API key**, a long string that identifies you on every request. Treat it like a password.
3. Make one request to the **Traffic Flow Segment Data** endpoint for a single point — pick a road you know, drop its latitude and longitude into the request, append your key.
4. Read the reply: it returns the segment's **current speed** and its **free-flow speed**. The gap between those two numbers is congestion, measured.

◆ EUREKA

A stretch of asphalt you chose just told you how fast it is moving, right now, in numbers. That is the whole business — you just did it once, by hand, for free.

HERE Traffic API

DEV-KEY
RENTED

Flow, incidents, and a congestion-severity score, sourced heavily from delivery and trucking fleets — which makes it strongest on freight corridors and interstates. A useful second opinion against TomTom.

`→ dashboard: freight-corridor speed overlay; cross-check against TomTom`

FIRST CONTACT — HERE

1. Register on the HERE platform and create a project to get your credentials.
2. Call the Traffic API's **flow** endpoint for a bounding box — a rectangle of map you define with two corner coordinates.
3. Read back per-segment speed and a **jam factor** (0 = clear, 10 = standstill). Compare a freight corridor here against the same road in TomTom.

Google Routes API

DEV-KEY
RENTED

Traffic-aware routing and travel times, priced per request with a monthly free credit. Broad urban coverage drawn from phone signals — strong where app users are dense.

`→ dashboard: live-versus-typical travel-time deltas per corridor`

FIRST CONTACT — GOOGLE

1. In the Google Cloud console, create a project and **enable the Routes API**, then generate an API key.
2. Send a **computeRoutes** request for a start and end point with traffic-aware routing switched on.
3. The reply's duration already accounts for live traffic; ask for it again without traffic and the difference is the corridor's current delay.

Mapbox Traffic Data

DEV-KEY

RENTED

Sells the raw speed dataset — both live speeds and typical-by-time-of-week — matched to OpenStreetMap. A free tier covers SDK use; the dataset itself is licensable on Rung Three. Of the four, it is the closest to data you can analyze rather than only display.

`→ dashboard: the typical-week baseline that live speeds are compared against`

FIRST CONTACT — MAPBOX

1. Create a Mapbox account; your **access token** is generated automatically on the account page.
2. Use a traffic-aware Directions request, or explore the typical-speed data, for a corridor you care about.
3. Note what makes this one different: the **typical** speeds give you a baseline for 'what this road normally does at 5pm on a Tuesday' — the reference every live reading is judged against.

◆ EUREKA

Four keys, four different windows onto the same street. You now hold every real-time input the dashboard's live layer needs — and it cost nothing but four sign-ups.

II.2 City-Cycle Benchmarks

Real-time feeds tell you this minute. Benchmarks tell you the *cycle* — the standardized, repeat-annually numbers that let you compare any city against any other on the same yardstick. This is the backbone of the ambition to understand every major metro's rhythm.

INRIX Global Traffic Scorecard

FREE
REPORT

The definitive city-cycle benchmark: three years of data across nearly a thousand cities, with per-driver hours lost, the dollar cost of congestion, the busiest corridors, and each metro's commute pattern. The full report is a free download. As a reference point, the 2025 edition put Chicago at 112 hours lost per driver, New York at 102, and the U.S. average at 49.

HOW TO USE IT

1. Download the current Scorecard report from `inrix.com/scorecard` (a short form-gate, then free).
2. Pull the table for your target city: hours lost, cost per driver, and the year-over-year change.
3. Keep last year's number beside this year's — the **direction** is usually the story, not the raw figure.

TomTom Traffic Index

FREE
REPORT

Live and historical congestion rankings by city, including Atlanta. Already on the newsdesk hub; here it pairs with INRIX as the second cross-city yardstick, so no single source carries a claim alone.

◆ EUREKA

Line Atlanta up against Chicago and New York on hours-lost, and you can speak to any metro's cycle with a number behind you. That is the ‘organic expertise’ goal beginning to compound — one comparison at a time.

II.3 Free Government Feeds

Public, unrestricted, and — unlike the commercial APIs — **ownable**. What the government publishes, you may keep and reprocess. These are the honest backbone of anything you want to build into a lasting archive.

State 511 / DOT feed

FREE
OWNABLE

Every state runs one; Georgia 511 is the local instance. Free real-time incidents, closures, and construction — no key, no storage restriction. Your home-turf edge, and a feed you can archive.

FHWA Highway Statistics

FREE
OWNABLE

Licensed drivers by age, vehicle-miles traveled, the authoritative annual tables — the national denominator behind any per-driver or per-mile claim.

Bureau of Transportation Statistics

FREE
OWNABLE

Broad federal mobility data for a deeper cut — freight flows, modal splits, long-run time series when a story needs history.

HOW TO PULL

1. Georgia 511 publishes a live incident feed — point the dashboard at it directly; it is the one real-time source on this rung you are free to store.
2. FHWA and BTS release tables you download once a year — keep them locally as the reference layer the live feeds get measured against.

II.4 Structural Context — The ‘Why’ Behind the Cycle

The layer that separates a real read from a surface one. Real-time says what is happening; structure says why a metro moves the way it does — where people live, where they work, and how they get between the two. All free, all ownable.

Census OnTheMap (LEHD / LODES)

FREE

OWNABLE

Home-to-work commute origin-destination, down to the census-block level. This is the structural reason a metro's peak looks the way it does — the map of who travels where, and when.

Census — data.census.gov (ACS)

FREE

OWNABLE

Commute times, mode share, and non-driver demographics by geography — the human layer beneath the traffic numbers, and the bridge to the newsdesk's ROADS-economy beat.

GTFS & GTFS-Realtime

FREE

OWNABLE

Open transit schedules plus live vehicle-position feeds. Transit ridership shifts explain much of a metro's road cycle — this is the multimodal half of the picture.

◆ EUREKA

With OnTheMap open beside a congestion number, you can now explain a jam, not just report it — ‘these blocks commute into that corridor at 8am.’ That is the moment reporting turns into analysis.

THE GITHUB LAB · FIELD MANUAL · SECTION

III

Rung Two

Agency Access — know it now, partner into it later

2
SOURCES

FREE
TO QUALIFY

STATUS
IS THE GATE

III.1 The Feeds Behind the Gate

These two are free — but not to individuals. They open to qualifying public agencies and partners, which means the cost is not money, it is standing. You cannot pull them solo today. They are on the ladder anyway, because knowing they exist shapes what you build toward: the DOT relationship the newsdesk and CZAR.tv already aim at is the same door these feeds sit behind.

NPMRDS via RITIS

AGENCY GATED

The FHWA probe-speed dataset for the entire national highway network, delivered through RITIS (the University of Maryland's CATT Lab). Free to qualifying public agencies and researchers. This is professional-grade segment speed history — the thing StreetLight and INRIX are often benchmarked against.

Waze for Cities

AGENCY GATED

A free, two-way partnership: agencies receive crowdsourced speeds, slowdowns, and incidents, and share their own closures back. Genuinely reciprocal, genuinely gated — agency standing is the key that opens it.

HOW TO REACH THEM

1. Note the qualification path now: both open to public agencies, MPOs, and named researchers — the exact standing the CZAR/DOT partnership track is built to earn.
2. Track the RITIS access terms and the Waze for Cities application; when a DOT relationship exists, these become available without new spend.
3. Until then, treat them as the 'why' behind the partnership strategy — real capability waiting on the other side of the door.

◆ EUREKA

You now know the two feeds that a single agency relationship unlocks — and why the DOT track is worth its patience. The door is named. That is progress even before it opens.

THE GITHUB LAB · FIELD MANUAL · SECTION

IV

Rung Three

The First Transaction — the milestone rung

$	OWN	1st
SPEND BEGINS	VS RENT	TXN

THRESHOLD · THE ACCOUNT OPENS HERE

This is where TM CZAR Television Network, Inc. makes its first transaction.

Not a shopping list — an earned graduation. The dashboard, built and proven on Rung One, is what justifies crossing this line. The account does not open because a tool is for sale; it opens because the free-tier work made the spend obvious.

IV.1 The Rule for the First Dollar

Carry one distinction through every card below: **paid access and paid ownership are not the same thing.** A higher TomTom, HERE, or Google tier still only rents you access — more calls, more speed, but the terms still forbid storing or redistributing the data. The transactions that move you toward a proprietary asset are the data-licensing ones. So the first dollar should point at something you can keep.

> *That is when the entity opens its bank account and starts to make its first transactions.*
>
> THE OPERATOR, ON WHY THE PAID RUNG EXISTS

IV.2 Buy Something You Own

Mapbox Traffic Data — license

PAID

OWNABLE

License the raw speed dataset outright, delivered to your own cloud storage — you keep it and reprocess it however you like. The clearest 'buy something you own' candidate for a first transaction, and a natural graduation from the Mapbox dev key on Rung One.

StreetLight InSight

PAID

OWNABLE

Licensed origin-destination, volumes, VMT, vehicle-hours of delay, and 15-minute intersection movements for nearly every road in North America. Enterprise pricing; the industry-standard pattern platform, most-used by public agencies.

HOW TO ACQUIRE — AND WHAT TO ASK

1. For Mapbox, request the Traffic Data license and confirm the delivery method (a cloud bucket you control) and the reprocessing rights — the whole point is that it lands somewhere you own.
2. For StreetLight, scope a single study first, not a platform subscription — one corridor or one metro's O-D — so the first spend is bounded and the value is provable.
3. On both, read the terms for the word that matters: can you **retain** the data after the term ends. That answer is the difference between an asset and a rental.

IV.3 Rent Broader Access

These are worth money too — but as access, not assets. Reach for them when the free tiers genuinely run out, and go in knowing the meter, not the deed, is what you are buying.

INRIX — analytics & real-time

PAID RENTED

Beyond the free Scorecard: real-time feeds and deeper analytics. Powerful, but access-only — the terms do not let you store or redistribute.

Replica

PAID RENTED

Enterprise activity- and travel-pattern modeling. Platform access, priced for agencies and large operators.

TomTom / HERE / Google — paid tiers

PAID RENTED

Higher call volumes and service levels on the very APIs from Rung One. More of what you were already renting — the storage and redistribution limits do not lift with the price.

IV.4 The North Star — Built, Not Bought

CMMD corridor capture

OWNED OWNABLE

Your own instrumented capture — the one data layer you fully own, and the endgame the entire ladder points at. Not a purchase; a build. Every rented feed above is a placeholder holding the shape of the thing until this exists.

◆ EUREKA

The first transaction is a milestone worth marking in the record: the day the entity stopped consuming for free and started acquiring on purpose — pointed, deliberately, at something it will keep.

THE GITHUB LAB · FIELD MANUAL · SECTION

V

The Build-Spec

The dashboard — the architecture you commission

6	**BEGINNER**	**PROOF**
STEPS	ASSUMED	OF SKILL

V.1 What the Dashboard Is

In plain terms: a single web page that pulls the Rung One feeds and shows, in one glance, what traffic is doing and how a city's cycle compares. Nothing exotic — a map with a live congestion layer, a panel of city-cycle numbers, and an incident list. It is small on purpose. Its job is not to be a product; its job is to prove you can wire the stack together, and to become the proprietary surface CMMD data eventually flows into.

V.2 The Architecture, in Plain Terms

Three parts, no jargon. A **front** — the page you see, with a map and panels. A **middle** — a small piece of code that holds your API keys and calls the feeds, so the keys never sit exposed in the page. A **store** — a place to keep the feeds you are allowed to keep (Georgia 511, the census and FHWA tables), so the dashboard has memory, not just a live view. You are not writing this from scratch; you are directing Claude Code to build it, one part at a time.

PART	PLAIN MEANING	BUILT FROM
Front	The page with the map and panels	A web page + a map library
Middle	The keeper of keys that calls the APIs	A small serverless function
Store	Memory for the ownable feeds	A simple database or files

V.3 The Build Order — and the Win at Each Step

Six steps, each a single instruction to hand the agent, each ending in something you can see. Build them in order; do not skip ahead, because each one proves the last one worked.

1. **Scaffold the page.** Direct Claude Code to stand up a blank web page with an empty map centered on Atlanta. Nothing live yet — just the canvas.
2. **Wire the first key.** Add the 'middle' that holds your TomTom key and calls the Flow endpoint for one segment. The page asks; the segment answers.
3. **Draw the live layer.** Color the roads on the map by current speed versus free-flow. Now the map *moves* — green where clear, red where jammed.
4. **Add the typical baseline.** Bring in Mapbox typical speeds so each road shows 'now versus normal,' not just now. This is the read no consumer app gives you.
5. **Build the city-cycle panel.** Drop in the INRIX and TomTom Index numbers so any metro can be compared on hours-lost beside the live map.
6. **Archive the ownable feed.** Point the 'store' at Georgia 511 and let it record incidents over time — the first data the dashboard owns rather than rents.

◆ EUREKA

Step three is the one that lands hardest: the first time the map colors itself from live data, a page you commissioned is showing the road breathing. That is the proof-of-skill the whole dashboard exists to demonstrate.

V.4 Why This Order Protects the Beginner

Each step is independently visible, so nothing is ever built on an unproven layer. If the map colors wrong at step three, you know the keys from step two are fine and the problem is only the drawing — the failure is always small and local. That is the beginner's real protection: not avoiding mistakes, but keeping every mistake tiny and obvious.

It also protects the ownership line. Notice that the only step that writes anything down — step six — is pointed at Georgia 511, an ownable feed. The rented APIs are shown live and never stored, which keeps the dashboard on the right side of every license by design, not by memory.

> ◆ EUREKA
>
> *When step six records its first incident, the dashboard stops being a live window and becomes an archive with a memory. That is the seed of the proprietary resource — the first row of data that is genuinely, keepably yours.*

THE GITHUB LAB · FIELD MANUAL · SECTION

VI

The Loop

How the build becomes public standing

3	**859SSCS1**	**MERGED**
MOVES	VAULT	= STANDING

VI.1 The Build Is Also a Credential

The dashboard is not only a tool — built in the open under the **859SSCS1** vault, it is a public, dated record of capability. The ledger's founding signal is exactly this: a non-technical founder became an open-source name by orchestrating agents and contributing upstream. The dashboard is that move, in your own domain.

◆ THE LOOP, APPLIED TO THE DASHBOARD

`STUDY the gap → IMPLEMENT via Claude Code → CERTIFY the published build → log it.`

STUDY: name what the dashboard is missing — a template, a data connector, a clean example. IMPLEMENT: direct the agent to build it against the repo's conventions. CERTIFY: publish it — a commit, a release, or a merged pull request is a dated, verifiable artifact of standing.

VI.2 The Build Register

Every certified step earns a line in the Build Register — the same discipline the newsdesk and the peer study close on. The register is what turns a pile of activity into a legible record of progress: what was done, when, and where the proof lives.

◆ EUREKA

The first published commit under 859SSCS1 is a credential with a date on it. From that moment the dashboard is not just something you made — it is something the record can point to.

THE GITHUB LAB · FIELD MANUAL · SECTION

VII

The Register

Build register, next actions, and the operator verbatim

5	BUILD	VERBATIM
NEXT ACTIONS	STATUS	PRESERVED

VII.1 Build Register — Next Actions

Five moves, in order, each drawn from a section above. Run them top to bottom and the ladder goes from documented to underway.

ID	ACTION	DETAIL
R-01	Claim the free keys	Sign up for TomTom, HERE, Google, and Mapbox dev tiers; make one call each (Section II.1).
R-02	Pull the benchmarks	Download the current INRIX Scorecard; line Atlanta up against two peer metros (II.2).
R-03	Scaffold the dashboard	Direct Claude Code through build steps one to three; get the live layer coloring (V.3).
R-04	Archive the ownable feed	Point the store at Georgia 511; record the first incident it owns (V.3, step six).
R-05	Certify the first line	Publish the build under github.com/859SSCS1; log it to the Build Register (VI).

◆ EUREKA

The day R-05 is logged, the manual has done its job: the free tier is mastered, the dashboard is real, and the first line of standing is on the record — all before a single dollar was spent.

VII.2 The Operator, Verbatim

Preserved as spoken across the working session that produced this manual — the intent, in the founder's own words.

> *"I want to have the widest range of data sources on traffic movement and patterns and real-time information with context sensitivity [than] anyone. I want to understand every major city's traffic cycles from this analysis and have that expertise be organic eventually through this experience."*

> *"I want a special section for the paid tier because that's when* TM CZAR Television Network, Inc. *opens up its bank account and starts to make its first transactions."*

> *"Want to master the free resources — and demonstrate my knowledge of how to use* API *via this* CZTVN *custom dashboard."*

> *"Assume beginner status and prepare to celebrate the eurekas of this implementation."*

END OF FIELD MANUAL · CZTVN-DATA-2026-0711 · GRADE A1

TM CZAR TELEVISION NETWORK, INC. · CZTVN · DJWZ CODING · DJWZ HOLDINGS LLC · SHANNON, GEORGIA

RESOURCE FACTS VERIFIED JULY 2026; ACCESS TERMS AND FIGURES REPORTED AS FOUND AND SUBJECT TO CHANGE.

VAULT: GITHUB.COM/859SSCS1 · LAYOUT PREPARED WITH CLAUDE (ANTHROPIC). STRATEGY AND FRAMING ARE THE OPERATOR'S.